

NANAKWA SHOAL BUOY 46181, Canada

WMO: 996520

Lat: 53.830N

Lon: 128.830W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.6	(c) -7.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 16.2	(k) -8.4	(l) 15.5	(m) -8.2	(n) 12.3	(o) 40	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.3	(c) 22.8	(d) N/A	(e) 21.1	(f) N/A	(g) 19.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.3	(b) 11.9	(c) 10.5	(e) -11.4	(f) 26.8	(g) 3.1	(h) 2.3	(i) -13.6	(j) 28.5	(k) -15.4	(l) 29.8	(m) -17.1	(n) 31.1	(o) -19.3	(p) 32.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	1.0	2.2	3.6	6.7	10.7	13.8	15.8	16.0	12.8	8.4	3.6	1.2
	DBStd	6.13	4.19	2.75	2.66	2.39	2.79	2.53	2.62	2.09	2.10	2.29	2.99	3.30
	HDD10.0	1350	279	217	198	104	24	1	0	0	3	60	193	271
	HDD18.3	3783	537	451	457	349	236	139	88	78	168	309	443	529
	CDD10.0	627	0	0	0	5	47	114	181	187	85	9	0	0
	CDD18.3	19	0	0	0	0	0	3	11	6	0	0	0	0
	CDH23.3	39	0	0	0	0	0	7	24	8	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	3	1	0	0	0	0
Wind	WSAvg	4.6	5.9	4.9	4.4	3.8	4.0	4.4	4.4	4.1	3.6	4.2	5.2	6.0
Precipitation	PrecAvg	2513	324	202	186	143	94	79	76	117	207	379	363	384
	PrecMax	3040	480	349	305	300	186	133	136	203	614	614	566	729
	PrecMin	2091	136	66	75	75	42	39	21	42	63	108	178	246
	PrecStd	289	87	87	61	62	47	28	31	53	134	109	101	129
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.1	9.8	16.7	20.5	24.4	25.9	24.3	20.3	14.3	10.4	8.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	6.9	6.8	8.5	13.8	18.3	21.3	23.4	21.9	18.1	13.1	9.0	6.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	5.9	6.0	7.5	11.7	16.5	19.2	21.5	20.5	16.7	12.2	8.0	5.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	4.9	5.2	6.6	10.1	14.8	17.6	19.7	19.2	15.5	11.3	7.0	4.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.9	3.3	3.9	4.6	5.1	4.6	4.4	4.3	3.6	3.1	2.9	2.7
		MCDBR	4.0	4.1	5.1	7.5	7.9	8.0	7.8	6.7	5.8	4.1	3.9	3.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.303	0.309	0.348	0.387	0.420	0.415	0.384	0.383	0.359	0.347	0.321	0.293
	taud		2.213	2.288	2.179	2.121	2.148	2.265	2.409	2.430	2.485	2.407	2.253	2.192
	Ebn at Noon		639	783	821	834	826	835	858	838	815	732	611	566
	Edh at Noon		63	81	113	137	141	127	108	100	84	73	61	53
All-Sky Solar Radiation	RadAvg		0.52	1.29	2.07	3.42	4.49	4.77	4.59	4.08	2.58	1.42	0.63	0.43
	RadStd		0.08	0.28	0.38	0.28	0.56	0.46	0.67	0.34	0.37	0.28	0.13	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page